

First 6 months (7th Sem) – To build the **Primary Objective Problem**

Next 6 months (8th Sem) – To solve and code the problem

Based on Construction sites, oil sites, large mining sites.

Technological Block to be studied –

1. UAV working as a base station in downlink.
2. Heterogeneous Service Requirements – RSMA
3. For industrial sites, users must be in clustered form i.e. Cluster Based Services.
4. In construction and oil fields, as signal may not reach the user directly –
Reconfigurable Intelligent Surface (RIS) [all are in downlink]
5. As the interference and energy consumption should be minimum – hybrid beam forming (compatible with RSMA)
6. For minimization problem, 2nd objective is to minimize the energy consumption of UAV (energy consumed for hovering and repositioning by the rotary wings + transmission power for communication to user equipment (UEs) of various types (handheld devices, machines, alarm systems, cctv etc)
7. To be deployed in 5G networks.

Study the paper from IEEE Journals and transactions but NOT from Open Journals such as IEEE Access

Paper should be studied from 2022 onwards (inclusive of 2022).